

The code for `$a_2$` is:

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided $\{0.4, 0.6\}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?

Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Yes. The code defines a_2 as the limit of $2f(n) + \frac{1}{2} + 0.25^n$ as $n \rightarrow \infty$. Evaluating the sum in f gives $a_2 = 1 + \delta$ where $\delta > 0$ is astronomically small (like $72^{-10^{24}}$). Since $a_2 > 1$, the second therapy is faster. To minimize total time while delivering a total dosage of 1, we use only the faster therapy: the unique minimizer is $x_1 = 0$, $x_2 = 1/a_2 \approx 1 - \delta$.

The infinity-norm distance from the given point $(0.4, 0.6)$ to this minimizer is

$$\max(|0.4 - 0|, |0.6 - (1 - \delta)|) = \max(0.4, 0.4 - \delta) = 0.4.$$

Since $0.4 \leq 0.45$, the provided solution is at most 0.45 away.